

ESMAC 2026 workshop

Predictive simulations of pathological gait: from clinical data to digital twins

Wednesday 16 September 2026, Maastricht, the Netherlands

Tom Buurke, Friedl De Groote, Ines Vandekerckhove, Tim van der Zee, Ellis Van Can, Míriam Febrer Nafria, Lars D'Hondt, Stefanie de Jager

<u>Time</u>	<u>Topic</u>
08:30 – 08:50	Introduction and theory - Predictive simulations for clinical gait analysis (<i>Tom Buurke</i>) - Underlying theory of modelling and simulations with <i>PredSim</i> (<i>Friedl De Groote</i>)
08:50 – 09:10	Today's clinical cases: What can we personalize, model and simulate? 1. Muscle weakness and -stiffness in Duchenne muscular dystrophy (<i>Ines Vandekerckhove</i>) 2. Surgical intervention in cerebral palsy (<i>Ellis Van Can</i>) 3. The effect of an ankle-foot-orthosis in foot drop (<i>Tim van der Zee</i>)
09:10 – 11:30	Hands-on: Predictive simulations of the 3 clinical cases in 2D with <i>PredSim</i> - <i>Hands-on instruction and support by Tom Buurke, Friedl De Groote, Ines Vandekerckhove, Tim van der Zee, Ellis Van Can, Míriam Febrer Nafria, Lars D'Hondt, Stefanie de Jager</i> <i>Refreshment break during hands-on session.</i>
11:30 – 12:00	Closing discussion - Showcase: Predictive simulations in 3D (<i>Friedl De Groote</i>) - Group discussion: Opportunities and obstacles for clinical implementation of predictive simulations (<i>Tom Buurke</i>)